

Technology Stack

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Postman (API Testing Client). No frontend UI — backend-only API project.	The project is a REST API backend. Postman serves as the client for testing and demonstrating all endpoints. A web/mobile frontend is planned as a future enhancement.
2	Backend / Server-Side	Node.js (v16+) with Express.js (v4.22)	Node.js provides a fast, event-driven JavaScript runtime. Express.js is a lightweight, unopinionated web framework ideal for building RESTful APIs. The team has existing expertise in JavaScript.
3	Database / Data Storage	MongoDB Atlas (Cloud NoSQL) with Mongoose ODM (v9.7)	MongoDB handles unstructured and flexible data (flashcard arrays, quiz arrays, study plan text) naturally. Mongoose provides schema validation and easy query methods. Atlas is cloud-hosted — no local database setup required.
4	Cloud / Hosting / Deployment	Local development server (localhost:5000). MongoDB Atlas for cloud database hosting.	Development and testing done locally using Nodemon dev server. MongoDB Atlas provides cloud-hosted database with built-in replication, backups, and 99.9% SLA.
5	Version Control & CI/CD	Git & GitHub	GitHub used for source code management, collaboration, and project submission. .gitignore configured to exclude node_modules and .env files.
6	Third-Party APIs / Other Tools	Google Gemini 2.5 Flash API via @google/generative-ai SDK (v0.24). JWT (jsonwebtoken v9.0) for authentication. bcryptjs (v3.0) for password hashing. Multer (v2.2) for file uploads.	Gemini API provides state-of-the-art generative AI for content generation (summaries, flashcards, quizzes, study plans). JWT enables stateless authentication. bcrypt ensures secure password storage. Multer handles multipart file uploads (.txt, .md, .pdf).